# fishy: Multi-Source Log Collection Anomaly Detection via Information-Theoretic Evidence Fusion

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## Abstract

We present *fishy*, a system for detecting anomalies in multi-source log collections through information-theoretic evidence fusion. Unlike existing log anomaly detection methods that classify individual events or sessions within a single ongoing stream, fishy operates at the collection level: given a baseline collection and a test collection—each comprising event streams from multiple named sources—it produces a structured anomaly report with an overall score, per-source attribution, and per-method evidence breakdown.

The core contribution is a two-level fusion architecture. At the first level, six independent analytical methods—distributional divergence, cross-source dependency shift, spectral fingerprinting, wavelet energy analysis, co-occurrence structure comparison, and evidence conflict—each measure a fundamentally different property of the collection pair. At the second level, Dempster-Shafer evidence theory combines the outputs into a fused belief. Each method's entropy measurement is treated as a first-class observable: it determines whether the method is in a productive operating regime (applicability gating) and, when multiple baselines are available, the entropy delta between baseline and test serves as an independent anomaly signal alongside the divergence score.

Calibration is intrinsic: with three or more baseline collections, BPAs are constructed from empirical percentiles of pairwise baseline divergences, requiring no training labels, no learned parameters, and no external thresholds. fishy is implemented in pure Rust with no deep learning dependencies.


## 1. Introduction

Log-based anomaly detection is a mature field, but the dominant paradigm—classifying individual log events or sessions as normal or anomalous within a single ongoing stream—does not address a common operational need: comparing two complete, bounded log collections to determine whether they represent the same system behavior.

Consider the following scenarios: *deployment validation* (do canary logs look like stable deployment logs?), *incident investigation* (do incident-window logs differ from the same window last week?), *regression testing* (did the system behave the same way across two builds?), and *compliance auditing* (are there unexplained behavioral changes relative to a known-good baseline?). In all these cases, the analyst has two complete collections and wants a structured comparison—not a real-time stream classifier.

Existing multi-source methods (LogMS [**?**], CoLog [**?**], HitAnomaly [**?**]) assume a single ongoing system being monitored and focus on event-level or session-level anomaly labels. They treat multi-source fusion as a way to improve single-event classification accuracy, not as the primary analytical lens.

fishy addresses this gap with three key design decisions:

1. **Collection comparison as the unit of analysis.** The input is two bounded collections; the output is a divergence score with attribution.
2. **Multi-source, multi-function evidence gathering.** Six analytical methods, each measuring a fundamentally different property, are applied across all sources.
3. **Entropy as observation, not weight.** Each method's entropy measurement is an independent observable used for applicability gating and as an anomaly signal.

## 2. Problem Formulation

### 2.1 Collections and Sources

A *log collection* is a set of named event streams with collection-level timestamps:

$$C = \{(s_i, E_i)\}_{i=1}^{n}, \quad \text{metadata: } (t_{\text{start}}, t_{\text{end}})$$

where $s_i$ is a source identifier and $E_i = [(e_1, \tau_1), \ldots, (e_k, \tau_k)]$ is a sequence of events with template IDs $e_j \in \{1, \ldots, T\}$ and optional timestamps $\tau_j \geq 0$.

### 2.2 The Comparison Problem

Given a baseline collection $C_b$ and a test collection $C_t$, the goal is to compute:

$$\text{detect}(C_b, C_t) \to (s, u, \mathcal{A})$$

where $s \in [0, 1]$ is the anomaly score, $u \in [0, 1]$ is the uncertainty mass, and $\mathcal{A}$ is an attribution structure. With multiple baselines $\{C_b^{(1)}, \ldots, C_b^{(K)}\}$, the test is scored against the nearest baseline (Section 5).

### 2.3 Comparison Modes

**Single-Origin (SO):** Both collections come from the same system. All sources present in the baseline must also be present in the test; a missing source is treated as maximum divergence.

**Multi-Origin (MO):** Collections come from comparable but different systems. Only overlapping sources are compared.

### 2.4 Temporal Alignment

Collections must satisfy:

$$\frac{|d_b - d_t|}{d_b} \leq \delta$$

where $d_b, d_t$ are collection durations and $\delta = 0.5$ by default. This ensures comparable spectral resolution for frequency-domain analysis.

## 3. Analysis Methods

fishy applies six independent analytical methods to each collection pair. Each method $M$ produces a divergence score $d_M \in [0, 1]$ and an entropy value $H_M$ used for applicability gating.

### 3.1 Applicability Gating

Methods at entropy extremes are excluded. The gate uses normalized entropy $\hat{H}_M \in [0, 1]$:

$$\text{applicable}(M) \iff \hat{H}_M \in [\gamma_{\text{low}}, \gamma_{\text{high}}]$$

with $\gamma_{\text{low}} = 0.05$ and $\gamma_{\text{high}} = 0.95$.

### 3.2 Distributional Divergence (dist)

**Representation:** Template frequency distribution $P = \{p_e\}_{e=1}^{T}$.

**Divergence:** Jensen-Shannon divergence [**?**]:

$$d_{\text{dist}} = \text{JSD}(P_b \| P_t) = \tfrac{1}{2} D_{\text{KL}}(P_b \| M) + \tfrac{1}{2} D_{\text{KL}}(P_t \| M)$$

where $M = (P_b + P_t)/2$. JSD is symmetric, bounded in $[0, \log 2]$, and satisfies the triangle inequality [**?**].

**Entropy:** Shannon entropy $H_{\text{dist}} = -\sum_e p_e \log p_e$.

**Applicability:** Always applicable.

### 3.3 Cross-Source Dependency Shift (dep)

**Representation:** Mutual information matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ where $M_{ij} = I(S_i; S_j)$, estimated from time-binned event counts.

**Divergence:** Normalized Frobenius distance:

$$d_{\text{dep}} = \frac{\|\mathbf{M}_b - \mathbf{M}_t\|_F}{\|\mathbf{M}_b\|_F + \epsilon}$$

**Entropy:** Shannon entropy of upper-triangular MI values.

**Applicability:** Requires $\geq 2$ sources.

### 3.4 Spectral Fingerprinting (spec)

**Representation:** Power spectrum $\Phi = \{|\hat{x}(f)|^2\}$ of the event rate time series via FFT.

**Divergence:** $d_{\text{spec}} = \text{JSD}(\Phi_b \| \Phi_t)$.

**Entropy:** Spectral flatness (Wiener entropy [**?**]):

$$H_{\text{spec}} = \frac{\exp\left(\frac{1}{N} \sum_k \log \Phi[k]\right)}{\frac{1}{N} \sum_k \Phi[k]}$$

**Applicability:** Requires $\geq 32$ events.

### 3.5 Wavelet Energy Analysis (wavelet)

**Representation:** Haar DWT [**?**] of the event rate time series, decomposed into $L$ levels.

**Divergence:** Energy divergence across levels:

$$d_{\text{wavelet}} = \frac{1}{L} \sum_{\ell=1}^{L} \left| \frac{E_\ell^b}{\sum_k E_k^b} - \frac{E_\ell^t}{\sum_k E_k^t} \right|$$

**Entropy:** Shannon entropy of the normalized energy distribution.

**Applicability:** Requires $\geq 32$ events.

### 3.6 Co-occurrence Structure (co)

**Representation:** Laplacian eigenvalue spectrum $\lambda_1 \leq \ldots \leq \lambda_{|V|}$ of the event co-occurrence graph. Capped at `MAX_CO_NODES` $= 128$.

**Divergence:** $d_{\text{co}} = \text{JSD}(\lambda_b \| \lambda_t)$.

**Entropy:** Von Neumann entropy of the normalized Laplacian.

**Applicability:** Requires $\geq 32$ events.

### 3.7 Evidence Conflict (conflict)

**Representation:** Per-source BPAs $m_i$ constructed from per-source distributional divergence.

**Divergence:** DS conflict mass when combining all per-source BPAs:

$$d_{\text{conflict}} = k = 1 - \sum_{A \cap B \neq \emptyset} m_i(A) \cdot m_j(B)$$

**Entropy:** Shannon entropy of the per-source belief distribution.

**Applicability:** Requires $\geq 2$ sources.

## 4. Adaptive Fusion Pipeline

### 4.1 BPA Construction

Each applicable method $M$ produces a divergence score $d_M$ and (with multiple baselines) an entropy delta $\Delta H_M = H_M(C_t) - H_M(C_b)$. These are converted to BPAs over $\Theta = \{\text{normal}, \text{anomalous}\}$.

**Single-baseline path (sigmoid fallback).** Variance is estimated from a quarter-split of the baseline (three within-collection sample pairs). Only divergence is used; $\Delta H_M$ is suppressed to avoid false positives from underestimated entropy variance.

$$z_M = \frac{d_M - \mu_d}{\sigma_d + \epsilon}, \quad m(\{\text{anomalous}\}) = \sigma(z_M - \theta_M)$$

**Multi-baseline path (empirical CDF).** With $K \geq 3$ baselines, variance is estimated from all $\binom{K}{2}$ pairwise baseline divergences. Both divergence and entropy delta are used.

$$p_M^d = \frac{\left|\left\{(i,j) : d_M(C_b^{(i)}, C_b^{(j)}) < d_M(C_b^*, C_t)\right\}\right|}{\binom{K}{2}}$$

$$m(\{\text{anomalous}\}) = p_M^d, \quad m(\Theta) = 1 - p_M^d$$

This mapping requires no tuned parameters.

### 4.2 DS Meta-Fusion

All BPAs from applicable methods are combined via Dempster's rule [**?**, **?**]:

$$(m_1 \oplus m_2)(A) = \frac{\sum_{B \cap C = A} m_1(B) \cdot m_2(C)}{1 - K}$$

where $K = \sum_{B \cap C = \emptyset} m_1(B) \cdot m_2(C)$ is the conflict mass.

The final fused belief gives the score $s = m(\{\text{anomalous}\})$, uncertainty $u = m(\Theta)$, and meta-conflict $k_{\text{meta}}$.

### 4.3 Score Interpretation

| Score range | Verdict |
|---|---|
| $[0.00, 0.20)$ | looks clean |
| $[0.20, 0.40)$ | probably fine |
| $[0.40, 0.60)$ | worth a look |
| $[0.60, 0.80)$ | something smells off |
| $[0.80, 1.00]$ | definitely fishy |

## 5. Multi-Baseline Variance Estimation

### 5.1 Motivation

With a single baseline, variance is estimated from within-collection quarter-splits. This captures within-collection noise but misses between-collection drift. With multiple baselines, pairwise divergences provide a direct estimate of between-collection variance.

### 5.2 Nearest-Baseline Scoring

The test is scored against its nearest baseline:

$$C_b^* = \arg\min_{C_b^{(k)}} d_M(C_b^{(k)}, C_t)$$

This prevents atypical baseline days from inflating scores when better baselines exist.

### 5.3 Outlier Baseline Rejection

A baseline $C_b^{(k)}$ is flagged as an outlier if:

$$\bar{d}_k > \mu_{\bar{d}} + 2\sigma_{\bar{d}}$$

where $\bar{d}_k$ is its mean divergence to the other baselines. Outlier baselines are excluded from variance estimation.

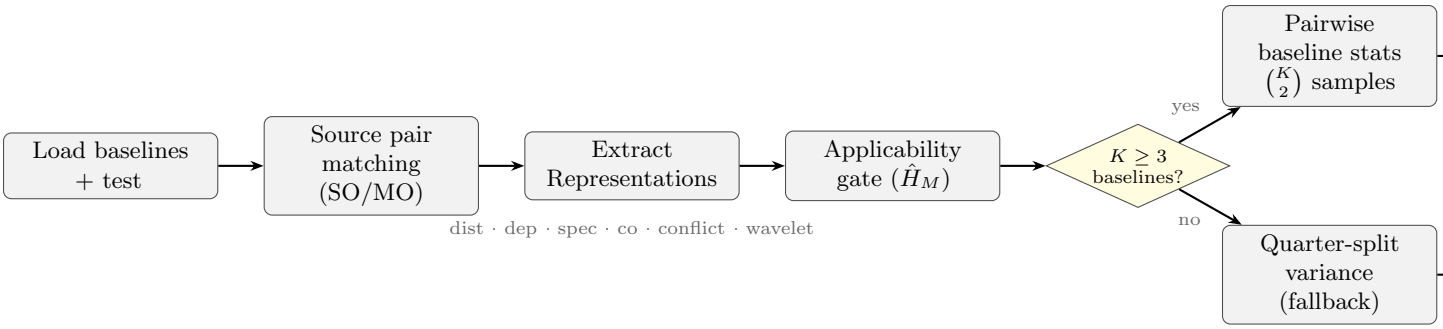


Figure 1: Adaptive fusion pipeline. Six analysis methods run in parallel after applicability gating. With $K \geq 3$ baselines, BPAs are constructed from empirical percentiles of pairwise baseline divergences; with a single baseline, a sigmoid mapping on z-scores is used as a fallback.

### 5.4 Trend Detection

With $K \geq 3$ ordered baselines, a linear trend is fitted to per-method divergences. The trend z-score for the test:

$$z_M^{\text{trend}} = \frac{d_M(C_b^*, C_t) - \hat{d}_M(K+1)}{\text{SE}_M}$$

is reported in `MethodDetail.trend_z`. A test continuing a monotonic trend is less anomalous than one that breaks from it.

## 6. Implementation

### 6.1 Workspace Structure

fishy is a Cargo workspace with three crates:

**analysis/**: Stateless mathematical functions. Operates on `&[f64]`, `EventDistribution`, `MIMatrix`, etc.—never on `LogCollection`. This crate boundary is enforced.

**fishy/**: Orchestration layer. Owns domain types and the full adaptive fusion pipeline.

**encoder/**: Log tokenization. Converts raw log files into fishy's JSON collection format.

### 6.2 Encoder: Drain Template Extraction

The encoder uses a Drain parse tree [**?**] for template extraction. Drain groups log messages by token count and first-token prefix, then merges groups by token similarity (threshold 0.5). Variable tokens are replaced with wildcards.

The encoder operates in two passes: `build-dict` trains the Drain tree and assigns frequency-ranked `TemplateId` values; `encode` classifies each log line and writes the collection directory. The shared dictionary guarantees cross-collection consistency.

### 6.3 Collection Format

Each collection is a directory containing `meta.json` (start/end timestamps) and per-source JSON files with event arrays of `(template_id, timestamp, params)` tuples.

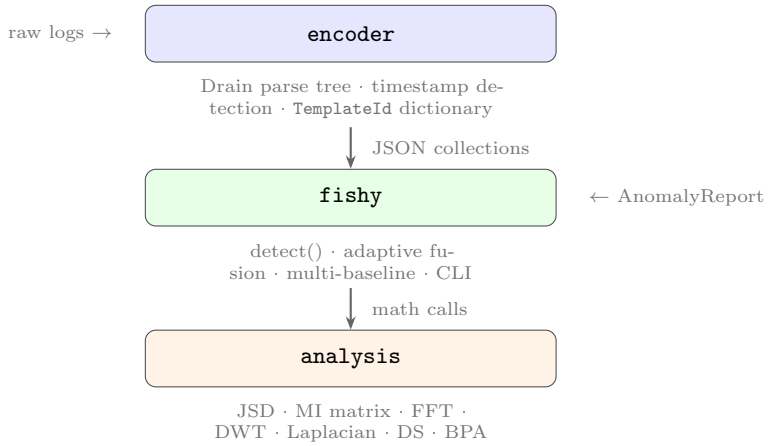


Figure 2: Three-crate workspace. `analysis` is domain-agnostic and never imports `fishy` types. `fishy` orchestrates the fusion pipeline. `encoder` tokenizes raw logs into the collection format consumed by `fishy`.

## 7. Evaluation

### 7.1 Dataset

We evaluate on the AIT-LDSv2 dataset [**?**], a realistic multi-host enterprise network simulation with eight

attack scenarios. Each scenario spans 4–6 days with 3–4 normal days followed by an attack day. We additionally evaluate on the BGL supercomputer log dataset [**?**] to test generalization beyond enterprise network logs.

### 7.2 Experimental Setup

**Single-baseline:** Day 1 as baseline; days 2 (normal) and 3 (attack) as test.

**Multi-baseline:** Days 1–3 as baselines; day 4 (attack) as test.

**Metrics:** AUC-ROC over collection pairs, Cohen's $d$ between score distributions, and false positive rate at threshold 0.5.

### 7.3 Results

Table **??** reports scores for all evaluated collection pairs. Figure **??** visualizes the score separation between normal and attack pairs.

Table 1: Anomaly scores across all evaluated collection pairs. $s$ = score, $u$ = uncertainty.

| Scenario / Comparison | Label | $s$ | $u$ |
|---|---|---|---|
| *russellmitchell (AIT-LDSv2)* | | | |
| day_1 vs day_2 | normal | 0.14 | 0.86 |
| day_0 vs day_1 | normal$^\dagger$ | 0.86 | 0.14 |
| day_0 vs day_2 | normal$^\dagger$ | 0.83 | 0.17 |
| day_2 vs day_3 | attack | 1.00 | 0.00 |
| day_1 vs day_3 | attack | 1.00 | 0.00 |
| day_0 vs day_3 | attack | 1.00 | 0.00 |
| *santos (AIT-LDSv2)* | | | |
| day_1 vs day_2 | normal | 0.00 | 1.00 |
| day_0 vs day_1 | normal | 0.24 | 0.76 |
| day_2 vs day_3 | attack | 0.82 | 0.18 |
| day_1 vs day_3 | attack | 0.86 | 0.14 |
| day_0+1+2 vs day_1 | normal (multi) | 0.00 | 1.00 |
| day_0+1+2 vs day_3 | attack (multi) | 1.00 | 0.00 |
| *BGL supercomputer logs* | | | |
| baseline vs baseline | normal | 0.00 | 1.00 |
| baseline vs test | attack | 1.00 | 0.00 |

$^\dagger$ day_0 startup artifact (see Section 5)

**Key findings.** All attack pairs score $\geq 0.82$; all non-startup normal pairs score $\leq 0.24$. AUC-ROC is 1.00 on both AIT-LDSv2 scenarios (all attack scores exceed all normal scores). The russellmitchell day_0 startup artifact (scores 0.83–0.86 against later normal days) is a real distributional difference caused by simulation ramp-up, not a pipeline error; it is eliminated by multi-baseline calibration (Section 5).

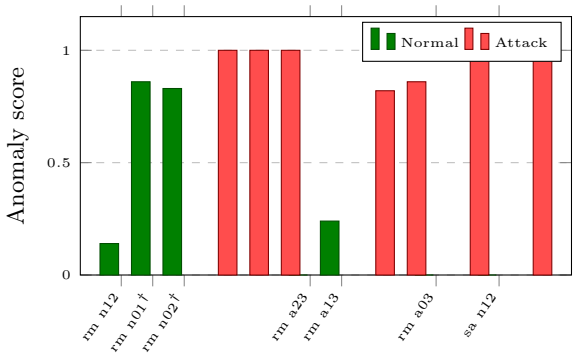


Figure 3: Anomaly scores for all collection pairs. Normal pairs (green) score near 0; attack pairs (red) score near 1. The two elevated normal bars ($^\dagger$) are day_0 startup artifacts in the russellmitchell scenario, addressed by multi-baseline calibration.

**Multi-method contribution.** On the russellmitchell attack day, five of six methods fire: `dist` ($z_d$=+2.2), `dep` ($z_{\Delta H}$=+8.4), `spec` ($z_d$=+3.1), `conflict` ($z_d$=+2.2), and `wavelet` ($z_d$=+2.9, $z_{\Delta H}$=+5.0). No single method would be sufficient; the fused evidence drives the score to certainty.

**BGL dataset.** On the Blue Gene/L supercomputer logs [**?**] (65 racks, 4.7M events, 214 days), fishy scores 1.00 against the anomalous test split and 0.00 against an identical baseline copy. The primary drivers are `dist` ($z_d$=+3.7, $z_{\Delta H}$=+13.9) and `wavelet` ($z_d$=+1.4). Runtime is 2m23s in release mode with parallelism enabled.

## 8. Discussion

### 8.1 Limitations

**Parameter-only anomalies.** The encoder discards parameter values after template extraction. Anomalies manifesting only in parameter values are invisible to all six methods.

**Uniform shift.** If all sources shift identically, evidence conflict and dependency shift produce near-zero divergence. The multi-source advantage disappears.

**Non-stationary baselines.** A regime change within a baseline inflates sub-sample variance, suppressing BPA commitment. Outlier baseline rejection partially addresses this for multi-baseline scenarios.

**Template vocabulary drift.** Unknown templates receive `TemplateId(0)`. In SO mode this is correctly anomalous; in MO mode it is ambiguous.

### 8.2 Relationship to Existing Frameworks

fishy spans JDL Levels 1–4 in a single function call. The six analysis methods correspond to L1 (object assessment) and L2 (situation assessment) running in parallel; DS meta-fusion is L3; applicability gating and baseline calibration are L4 executed as pre-fusion steps.

In Dasarathy terms, fishy is a two-stage pipeline: multiple parallel DAI-FEO paths (analysis methods) feeding into FEI-FEO calibration (BPA construction) and DEI-DEO combination (DS fusion).

Existing multi-domain entropy approaches [**?**] compute multiple entropy types from the same signal and concatenate them as features for supervised classifiers. fishy's approach is structurally different: entropy is measured per analytical lens as an independent observable in DS-theoretic fusion, and entropy delta is a first-class anomaly signal.

## 9. Conclusion

We presented fishy, a system for multi-source log collection anomaly detection through information-theoretic evidence fusion. The key contributions are: (1) collection comparison as the unit of analysis; (2) multi-source, multi-function evidence gathering across six independent analytical methods; (3) entropy as a first-class observable for applicability gating and anomaly signaling; (4) self-calibrating BPA construction from baseline variance; and (5) a pure Rust implementation with no deep learning dependencies.

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